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MASTER'S THESIS

From Internal Representations to Output Distributions: Improving Reliability, Control, and Training Efficiency in Large Language Models

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МАГИСТЕРСКАЯ ДИССЕРТАЦИЯ

От внутренних представлений к выходным распределениям: повышение надёжности, управляемости и эффективности обучения больших языковых моделей

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From Internal Representations to Output Distributions: Improving Reliability, Control, and Training Efficiency in Large Language Models

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ABSTRACT

Large language models (LLMs) have demonstrated remarkable capabilities across diverse tasks, yet their deployment faces three critical challenges: unreliable outputs that may contain errors or hallucinations, limited control over generation behavior, and inefficient training procedures that waste computational resources on redundant data. This thesis addresses these challenges through a unified approach that leverages signals from the model's forward pass—both internal representations and output distributions—to improve reliability, control, and training efficiency.

We present three interconnected studies. First, we investigate uncertainty estimation by analyzing the relationship between token-level entropy and answer correctness. Our experiments on MMLU-Pro across four models (Phi-4, Mistral-Small-24B, Qwen-1.5B, Qwen-72B) demonstrate that single-token answer entropy achieves ROC-AUC scores up to 0.83 for predicting incorrect answers in knowledge-dependent domains, while Model-as-Judge approaches perform near-randomly. Second, we explore the geometric structure of multilingual representations in LLM latent space, discovering that language-specific directions can be identified through PCA with 96–100% classification accuracy across four language pairs. By projecting out the source language component from hidden states, we reduce distributional divergence by up to 55% and achieve 63–95% reduction in Code-Switching Index in generated text, without requiring additional training. Third, we develop a complexity-aware fine-tuning pipeline that uses entropy to stratify training data and applies targeted interventions: direct supervised fine-tuning for regular examples and chain-of-thought distillation for complex ones. This approach achieves accuracy improvements from 0.39 to 0.52 (Qwen-3B) and 0.51 to 0.64 (Phi-4-Mini) on MMLU-Pro while using 79–82% less training data, with results generalizing to GSM8K and MedMCQA datasets.

Together, these contributions demonstrate that accessible signals from a single forward pass provide valuable information for addressing fundamental challenges in LLM deployment. The methods developed require no architectural modifications or expensive retraining, making them practical for real-world applications.

Keywords: large language models, uncertainty estimation, latent space steering, complexity-aware fine-tuning, internal representations

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От внутренних представлений к выходным распределениям: повышение надёжности, управляемости и эффективности обучения больших языковых моделей

Андрей Гончаров

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Аннотация

Большие языковые модели (LLM) продемонстрировали выдающиеся способности в решении разнообразных задач, однако их внедрение сталкивается с тремя критическими проблемами: ненадёжные выходные данные, которые могут содержать ошибки или галлюцинации, ограниченный контроль над генерацией и неэффективные процедуры обучения, расходующие вычислительные ресурсы на избыточные данные. Данная диссертация решает эти проблемы с помощью единого подхода, использующего сигналы прямого прохода модели — как внутренние представления, так и выходные распределения — для повышения надёжности, управляемости и эффективности обучения.

Мы представляем три взаимосвязанных исследования. Во-первых, мы исследуем оценку неопределённости путём анализа связи между энтропией на уровне токенов и правильностью ответов. Наши эксперименты на MMLU-Pro с четырьмя моделями (Phi-4, Mistral-Small-24B, Qwen-1.5B, Qwen-72B) показывают, что энтропия ответного токена достигает ROC-AUC до 0.83 для предсказания неправильных ответов в доменах, зависящих от знаний, тогда как подходы Model-as-Judge работают почти случайно. Во-вторых, мы исследуем геометрическую структуру мультиязычных представлений в латентном пространстве LLM, обнаруживая, что языковые направления могут быть идентифицированы с помощью PCA с точностью классификации 96–100% для четырёх языковых пар. Проецируя компоненту исходного языка из скрытых состояний, мы уменьшаем дистрибуционную дивергенцию до 55% и достигаем снижения индекса переключения кода на 63–95% в сгенерированном тексте без дополнительного обучения. В-третьих, мы разрабатываем метод дообучения с учётом сложности, который использует энтропию для стратификации обучающих данных и применяет целевые интервенции: стандартное дообучение для обычных примеров и дистилляцию цепочки рассуждений для сложных. Этот подход повышает точность с 0.39 до 0.52 (Qwen-3B) и с 0.51 до 0.64 (Phi-4-Mini) на MMLU-Pro при использовании на 79–82% меньше обучающих данных, с обобщением результатов на датасеты GSM8K и MedMCQA.

В совокупности эти вклады демонстрируют, что доступные сигналы одного прямого прохода предоставляют ценную информацию для решения фундаментальных проблем внедрения LLM. Разработанные методы не требуют архитектурных модификаций или дорогостоящего переобучения, что делает их практичными для реальных приложений.

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Chapter 1
Introduction

Note on the use of AI tools: *In some portions of this document (20% - 30% of the entire text), Artificial Intelligence assistants, particularly Generative AI, have been used to improve, rephrase, and summarize the content. The technologies used include Claude Code.*

Large language models (LLMs) have emerged as one of the most transformative technologies in artificial intelligence, demonstrating remarkable capabilities across diverse domains including natural language understanding, code generation, mathematical reasoning, and creative writing [5, 36, 44]. These models, trained on vast corpora of text data, have shown an unprecedented ability to generalize across tasks and generate coherent, contextually appropriate responses. However, despite their impressive performance, the deployment of LLMs in real-world applications faces several fundamental challenges that limit their reliability, controllability, and efficiency.

Relevance. The widespread adoption of LLMs in critical applications—from healthcare diagnostics to legal document analysis, from educational tutoring to scientific research assistance—demands that these systems be trustworthy and predictable. Yet current LLMs exhibit three interconnected limitations that undermine their practical utility:

1. **Unreliable outputs**: LLMs frequently produce incorrect or fabricated information with high apparent confidence, a phenomenon known as hallucination [21]. Users cannot easily distinguish between reliable and unreliable model outputs, making it difficult to trust LLM-generated content without extensive verification.

2. **Limited behavioral control**: Once trained, LLMs exhibit fixed behavioral patterns that may not align with user requirements. For multilingual models, this manifests as unwanted code-switching, where models unexpectedly switch languages mid-response [48]. More broadly, fine-grained control over generation behavior remains challenging without expensive retraining.

3. **Inefficient training procedures**: Fine-tuning LLMs on new tasks or domains requires substantial computational resources. Current approaches often treat all training examples equally, ignoring the varying complexity and informativeness of different data points, leading to wasted computation on redundant or trivial examples.

These challenges are not merely academic concerns—they represent practical barriers to LLM adoption in high-stakes applications where errors carry significant consequences.

Main purpose of the research. This thesis investigates a unified approach to addressing these three challenges by leveraging signals that are readily available from a single forward pass through the model. Specifically, we exploit two types of signals: (1) *output distributions*, the probability distributions over tokens that models produce at each generation step, and (2) *internal representations*, the high-dimensional activation patterns in the model's hidden layers. Our central hypothesis is that these signals encode rich information about model behavior, uncertainty, and input characteristics that can be extracted and utilized without requiring architectural modifications or additional training.

The research addresses three concrete questions:

1. **RQ1**: How can patterns in token-level output distributions be used to characterize model uncertainty and predict answer correctness?
2. **RQ2**: What geometric structure exists in the latent representations of multilingual LLMs, and how can this structure be exploited to control language-specific behaviors?
3. **RQ3**: How can output distribution characteristics guide the selection and processing of training data to improve fine-tuning efficiency?

Scientific novelty. This thesis makes several novel contributions to the field of LLM research:

1. We present a systematic analysis of token-wise entropy as an uncertainty indicator, demonstrating that single-token answer entropy effectively predicts answer correctness in knowledge-dependent domains (ROC-AUC up to 0.83) while revealing that entropy primarily captures knowledge uncertainty rather than reasoning difficulty. This work was recognized with the **Best Paper Award at AINL 2025**.
2. We discover and characterize language-specific directions in LLM latent space that can be identified through PCA with 96–100% classification accuracy across four language pairs. We demonstrate that projecting out the source language component from hidden states reduces code-switching by 63–95% in generated text without any additional training, providing a new mechanism for behavioral control.
3. We develop a complexity-aware fine-tuning pipeline that combines entropy-based data stratification with targeted training interventions (direct SFT for regular examples, chain-of-thought distillation for hard examples), achieving accuracy improvements while reducing training data requirements by 79–82%, validated across three datasets and three model architectures.
4. We establish a unifying framework showing how signals from the forward pass—whether from output distributions or internal representations—can address diverse challenges in LLM deployment, suggesting broader applicability of these techniques.

Statements for defense.

1. Single-token answer entropy provides an effective signal for predicting LLM answer correctness in knowledge-dependent domains, achieving ROC-AUC scores up to 0.83 on the MMLU-Pro benchmark, while Model-as-Judge approaches perform near-randomly (ROC-AUC $\approx$ 0.49–0.57).
2. Language-specific directions exist in the latent space of multilingual LLMs and can be reliably identified through PCA with 96–100% classification accuracy across four language pairs. Projecting out the source language component from hidden states reduces code-switching by 63–95% as measured by Code-Switching Index in generated text.
3. Complexity-aware fine-tuning that stratifies training data by entropy and applies appropriate training strategies (direct SFT for regular examples, chain-of-thought distillation for hard ones) improves model accuracy while requiring 79–82% less training data than full distillation, generalizing across three datasets and three model architectures.
4. Signals from a single forward pass through an LLM—both output distributions and internal representations—encode sufficient information to address fundamental challenges in reliability, control, and training efficiency without requiring architectural modifications.

Chapter 2
Author contribution

This thesis is based on three research papers that have been submitted to peer-reviewed venues. Below we describe the author's specific contributions to each work.

Paper 1: When an LLM is Apprehensive About Its Answers — And When Its Uncertainty is Justified

Authors: P. Sychev, A. Goncharov, D. Vyazhev, E. Khalafyan, A. Zaytsev

Venue: Proceedings of AINL 2025 (**Best Paper Award**). arXiv preprint arXiv:2503.01688.

Summary: This paper investigates uncertainty estimation in large language models by analyzing token-level entropy patterns across different question topics and reasoning requirements. We demonstrate that entropy-based features can predict answer correctness with ROC-AUC up to 0.83 in knowledge-dependent domains, while Model-as-Judge (MASJ) performs near-randomly. We also reveal internal biases in the MMLU-Pro dataset regarding question complexity.

Author contribution: The thesis author (A. Goncharov) contributed to the experimental design, implementation of entropy analysis methods, data collection and processing, and analysis of results. The author participated in developing the methodology for extracting and analyzing token-wise entropy patterns and contributed to writing the experimental sections of the paper.

Paper 2: Complexity-Aware Fine-Tuning

Authors: A. Goncharov, D. Vyazhev, P. Sychev, E. Khalafyan, A. Zaytsev

Venue: Proceedings of EACL 2026. arXiv preprint arXiv:2506.21220.

Summary: This paper presents a complexity-aware fine-tuning pipeline that uses single-token answer entropy to stratify training data by complexity and applies targeted training interventions: standard SFT for regular examples and chain-of-thought distillation for hard examples. The approach achieves significant accuracy improvements while using 79–82% less training data than full distillation baselines, validated across three datasets (MMLU-Pro, GSM8K, MedMCQA).

Author contribution: The thesis author (A. Goncharov) is the lead author and primary contributor. The author designed the overall methodology, implemented the entropy-based data stratification system, developed the training pipeline combining SFT and chain-of-thought distillation, conducted all experiments, analyzed results, and wrote the majority of the paper.

Paper 3: Language Steering in Latent Space to Mitigate Unintended Code-Switching

Authors: A. Goncharov, N. Kondusov, A. Zaytsev

Venue: Proceedings of AINL 2026. arXiv preprint arXiv:2510.13849.

Summary: This paper explores the geometric structure of multilingual representations in LLM latent space. We discover that language-specific directions can be identified through

PCA with 96–100% classification accuracy across four language pairs (English–Chinese, English–Spanish, English–Russian, English–Hindi). Steering activations via projection onto these directions reduces distributional divergence by up to 55% and achieves 63–99% reduction in Code-Switching Index in generated text.

Author contribution: The thesis author (A. Goncharov) is the lead author and primary contributor. The author conceived the idea of using PCA to identify language directions, designed and implemented the steering methodology, conducted experiments across multiple model architectures and language pairs, performed the analysis of results, and wrote the majority of the paper.

Chapter 3
List of publications

1. Sychev P., Goncharov A., Vyazhev D., Khalafyan E., Zaytsev A. When an LLM is Apprehensive About Its Answers — And When Its Uncertainty is Justified // Proceedings of AINL 2025 (**Best Paper Award**). arXiv preprint arXiv:2503.01688.

2. Goncharov A., Vyazhev D., Sychev P., Khalafyan E., Zaytsev A. Complexity-Aware Fine-Tuning // Proceedings of EACL 2026. arXiv preprint arXiv:2506.21220.

3. Goncharov A., Kondusov N., Zaytsev A. Language Steering in Latent Space to Mitigate Unintended Code-Switching // Proceedings of AINL 2026. arXiv preprint arXiv:2510.13849.

Chapter 4
Literature review

This chapter surveys the research landscape relevant to the three main themes of this thesis: uncertainty estimation in language models, interpretability and control of internal representations, and efficient training methodologies.

4.1 Uncertainty Estimation in Large Language Models

The problem of uncertainty quantification in neural networks has a long history [14], but the advent of large language models has introduced new challenges and opportunities. Traditional approaches to uncertainty estimation, such as Monte Carlo dropout [14] and deep ensembles [28], require multiple forward passes and are computationally expensive for large models.

Recent work has explored more efficient alternatives. Confidence calibration methods attempt to align model confidence scores with actual accuracy [15], but LLMs often exhibit poor calibration, particularly on out-of-distribution inputs [23]. Verbalized confidence, where models are prompted to express uncertainty in natural language [24, 30], offers an interesting alternative but relies on the model's ability to accurately self-assess.

Token-level probability analysis has emerged as a promising direction. Kuhn et al. [27] introduced semantic entropy, which clusters generated sequences by meaning before computing entropy, addressing the challenge of semantically equivalent but lexically different outputs. Malinin and Gales [32] proposed using the entropy of the predictive distribution as an uncertainty measure for sequence generation tasks.

The relationship between entropy patterns during generation and answer correctness remains underexplored. While prior work has focused on aggregate measures or first-token probabilities, our research investigates how token-wise entropy dynamics throughout the generation process correlate with model reliability.

4.2 Internal Representations and Interpretability

Understanding the internal representations of neural networks has been a central goal of interpretability research [35, 10]. For language models, this involves understanding what information is encoded in activation patterns and how this information is processed across layers.

Probing classifiers have been widely used to identify what linguistic and semantic information is encoded in model representations [2, 8]. These studies have revealed that different layers capture different types of information, with lower layers encoding syntactic features and higher layers encoding more semantic content [20].

More recently, researchers have discovered that model representations exhibit meaningful geometric structure. Linear directions in activation space have been found to correspond to interpretable concepts [33, 38]. This has enabled activation steering approaches, where model behavior can be modified by adding vectors to intermediate activations [45, 29]. Zou et al. [54] introduced representation engineering, using PCA on contrastive prompt pairs to identify concept directions (e.g., honesty, harmlessness) and steering activations along them. In parallel, concept erasure meth-

ods like iterative null-space projection (INLP) [42] and LEACE [3] remove specific attributes from representations by projecting onto their null space.

For multilingual models, the question of how different languages are represented has received significant attention. Work on mBERT and XLM-R revealed that language identity concentrates in few principal components of hidden states, separable from semantics in parallel texts [9, 6]. Yang et al. [52] removed these components for language-agnostic embeddings. Wendler et al. [47] extended this to decoder-only LLMs, revealing interpretable geometric structure in internal representations. While prior work focused on analysis, our work presents an early exploration of exploiting this structure for active generation control, using language identity as a practical test case.

Our work extends these findings by demonstrating that language-specific directions can be identified through simple PCA analysis and used for inference-time steering without additional training.

4.3 Code-Switching in Multilingual Models

Code-switching refers to the alternation between languages within a single conversation or text. While natural in multilingual human communication, unintended code-switching in LLM outputs is generally undesirable as it reduces output quality and user experience [11].

Prior approaches to controlling code-switching have primarily relied on training-time interventions. Language-specific fine-tuning can reduce code-switching but requires separate models or adapters for each language [40]. Constrained decoding methods can force outputs to remain in a target language but may degrade generation quality [41].

The discovery that language identity is encoded as linear directions in latent space [47] opens new possibilities for inference-time control. However, practical methods for exploiting this structure to reduce code-switching have not been thoroughly explored. Our work addresses this gap by developing and evaluating activation steering methods specifically designed for language control.

4.4 Efficient Fine-Tuning and Data Selection

Fine-tuning large language models is computationally expensive, motivating research into more efficient approaches. Parameter-efficient fine-tuning methods such as LoRA [19] and adapters [17] reduce the number of trainable parameters but do not address data efficiency.

Data selection and curriculum learning aim to improve training efficiency by carefully choosing which examples to train on. Influence functions [26] can identify the most impactful training examples but are expensive to compute for large models. Active learning approaches [43] select informative examples iteratively but require multiple training rounds.

Recent work has explored using model predictions to guide data selection. Self-training and self-distillation methods [13, 50] use model outputs as targets for further training. Chain-of-thought distillation [16, 31, 18] transfers reasoning capabilities from larger to smaller models by training on generated rationales.

The idea of stratifying training data by difficulty has been explored in curriculum learning [4, 25], where models are trained on progressively harder examples. The LIMA approach [53] demonstrated that small, high-quality datasets suffice for alignment, showing that data curation can substitute for data volume. Fadeeva et al. [12] provided comprehensive comparisons of white-box and black-box complexity estimation methods, presenting promising results for entropy-based approaches. However, existing methods do not leverage the insight that examples of different complexity may benefit from different training strategies. Our work proposes using entropy as

an automatic measure of example complexity and demonstrates that different complexity levels benefit from different training interventions.

4.5 Summary and Research Gaps

The literature reveals several research gaps that this thesis addresses:

1. While uncertainty estimation methods exist, the relationship between token-wise entropy dynamics and answer correctness has not been systematically studied.
2. Although language-specific directions in latent space have been identified, practical methods for using them to control code-switching at inference time are lacking.
3. Existing data selection methods do not leverage the insight that examples of different complexity may benefit from different training strategies.

This thesis addresses these gaps through three interconnected studies that exploit signals from the model's forward pass to improve reliability, control, and efficiency.

Chapter 5
Problem statement

This chapter formally defines the research problems addressed in this thesis and establishes the notation used throughout.

5.1 Preliminaries and Notation

Let $\mathcal{M}$ denote an autoregressive language model with vocabulary $\mathcal{V}$ of size $|\mathcal{V}|$. Given an input sequence $x = (x_1, \ldots, x_n)$, the model produces a probability distribution over the next token:

$$p(x_{n+1}|x_1, \ldots, x_n) = \text{softmax}(f_\theta(x_1, \ldots, x_n)) \tag{5.1}$$

where $f_\theta : \mathcal{V}^n \to \mathbb{R}^{|\mathcal{V}|}$ represents the model's logit function parameterized by θ.

For a generated sequence $y = (y_1, \ldots, y_m)$, we define the token-wise entropy at position t as:

$$H_t = -\sum_{v \in \mathcal{V}} p(v|x, y_1, \ldots, y_{t-1}) \log p(v|x, y_1, \ldots, y_{t-1}) \tag{5.2}$$

Let $h_l^{(t)} \in \mathbb{R}^d$ denote the hidden state at layer l and position t, where d is the hidden dimension of the model.

5.2 Problem 1: Uncertainty Estimation via Output Distributions

Definition 5.1 (Answer Correctness Prediction) *Given a question q and a model-generated answer a, the answer correctness prediction problem is to estimate $P(\text{correct}|q, a, \mathcal{M})$ using only information available from a single forward pass through $\mathcal{M}$.*

The goal is to identify features derived from the output distribution that correlate with answer correctness. Specifically, we investigate:

1. The sequence of token-wise entropies $(H_1, H_2, \ldots, H_m)$ during answer generation
2. Aggregate statistics including mean entropy, maximum entropy, minimum entropy, and their ratios
3. The predictive power of these features for distinguishing correct from incorrect answers

Objective: Develop entropy-based features that achieve high ROC-AUC for predicting answer correctness, improving upon baseline confidence measures.

5.3 Problem 2: Language Control via Latent Space Steering

Definition 5.2 (Language Direction) *A language direction $v_\ell \in \mathbb{R}^d$ for language ℓ is a unit vector in the model's hidden state space such that the projection of hidden states onto v_ℓ is predictive of whether the model is generating text in language ℓ.*

Definition 5.3 (Code-Switching) *Code-switching occurs when a model, prompted to respond in language ℓ_1, produces tokens that belong to a different language $\ell_2 \neq \ell_1$.*

The language steering problem consists of two sub-problems:

Sub-problem 2a (Direction Identification): Given a multilingual model $\mathcal{M}$ and a set of languages $\mathcal{L}$, identify language directions $\{v_\ell\}_{\ell \in \mathcal{L}}$ from hidden states using unsupervised methods.

Sub-problem 2b (Activation Steering): Given identified language directions, modify the model's hidden states during inference to reduce code-switching by projecting out the source language component:

$$\tilde{h}_l^{(t)} = h_l^{(t)} - s \left(h_l^{(t)} \cdot v^{(l)} \right) v^{(l)} \tag{5.3}$$

where $s \in \mathbb{R}$ controls intervention strength. When $s = 1$, this is an exact orthogonal projection that removes the language component; values $|s| > 1$ actively push representations away from the source language direction.

Objective: Achieve high language classification accuracy ($> 95\%$) for direction identification and significant reduction in code-switching ($> 50\%$ reduction in KL divergence; $> 60\%$ reduction in Code-Switching Index in generated text).

5.4 Problem 3: Complexity-Aware Fine-Tuning

Definition 5.4 (Example Complexity) *The complexity of a training example (x, y) with respect to model $\mathcal{M}$ is measured by the entropy of the model's output distribution when generating y given x:*

$$C(x, y; \mathcal{M}) = \frac{1}{|y|} \sum_{t=1}^{|y|} H_t \tag{5.4}$$

The complexity-aware fine-tuning problem is to design a training procedure that:

1. Stratifies training data $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$ into complexity tiers based on $C(x_i, y_i; \mathcal{M})$
2. Applies appropriate training interventions for each tier:
   - Low complexity (easy examples): Direct supervised fine-tuning (SFT)
   - High complexity (hard examples): Chain-of-thought distillation from a teacher model
3. Achieves higher accuracy with less training data compared to uniform training

Formally, let $\mathcal{D}_{\text{easy}} = \{(x, y) \in \mathcal{D} : C(x, y; \mathcal{M}) < \tau\}$ and $\mathcal{D}_{\text{hard}} = \mathcal{D} \setminus \mathcal{D}_{\text{easy}}$ for some threshold τ. The training objective is:

$$\mathcal{L} = \sum_{(x,y) \in \mathcal{D}_{\text{easy}}} \mathcal{L}_{\text{SFT}}(x, y) + \sum_{(x,y) \in \mathcal{D}_{\text{hard}}} \mathcal{L}_{\text{distill}}(x, y, \mathcal{M}_{\text{teacher}}) \tag{5.5}$$

Objective: Achieve accuracy improvements over baseline SFT while using significantly less training data ($> 50\%$ reduction in data requirements).

5.5 Unifying Theme

All three problems share a common approach: extracting and utilizing information from a single forward pass through the model. This information takes two forms:

1. **Output distributions**: The probability distributions over tokens (Problems 1 and 3)
2. **Internal representations**: The hidden states at intermediate layers (Problem 2)

The unifying hypothesis is that these signals encode rich information about model behavior, uncertainty, and input characteristics that can be exploited without architectural modifications or additional training. This thesis validates this hypothesis through empirical investigation of all three problems.

Chapter 6
Methodology

This chapter describes the theoretical approaches, methods, and algorithms used in each of the three studies comprising this thesis.

6.1 Study 1: Uncertainty Estimation via Token-Wise Entropy

Entropy Extraction

For each generated token y_t in a model's response, we extract the full probability distribution over the vocabulary and compute the Shannon entropy:

$$H_t = -\sum_{v \in \mathcal{V}} p_t(v) \log_2 p_t(v) \tag{6.1}$$

where $p_t(v) = p(v|x, y_1, \ldots, y_{t-1})$ is the probability assigned to token v at position t.

Uncertainty Estimation Approaches

We consider two approaches to uncertainty estimation: entropy-based and Model-as-Judge (MASJ).

Single-token answer entropy. For MMLU-Pro questions with multiple-choice answers, the expected answer is one or two tokens long. We prompt the model to answer directly (without chain-of-thought) and compute the token-wise entropy of the answer token from the model's output logits. This serves as our primary complexity metric, as response uncertainty is a natural predictor of question complexity.

Chain-of-thought entropy variants. We also analyze entropy when the model produces a chain-of-thought response. From the CoT response, we compute several aggregation features:

1. **CoT Mean**: mean entropy across all tokens in the reasoning chain
2. **CoT Max**: maximum entropy in the reasoning chain
3. **Sequence-aggregation methods**: mean/max aggregated at the word level
4. **Marginal difference**: mean difference between top-2 token probabilities

Model-as-Judge (MASJ) Approach

As a complementary approach, we use a large auxiliary LLM (Mistral-Large-Instruct-2411, 123B parameters) to estimate question properties:

- **Reasoning requirement**: whether the question requires reasoning and how many steps (low, medium, high)
- **Education level**: estimated difficulty as educational level (high school, undergraduate, graduate, postgraduate)

The MASJ scores are obtained via specific prompts and filtered using an MT-Bench-style quality assessment: instances where the judge assigns a score below 8 on a scale of 1 to 10 are regenerated.

Evaluation Protocol

We compare uncertainty estimates with ground-truth correctness labels using ROC-AUC, stratified by subject domain (14 topics), reasoning requirement (low/medium/high), and model architecture. Answers with incorrect formatting (less than 5% of questions) are discarded.

6.2 Study 2: Language Steering in Latent Space

Hidden State Extraction

For a multilingual model $\mathcal{M}$ and a parallel corpus $\mathcal{D} = \{(s_i, k_i)\}_{i=1}^{N}$ where s_i represents semantically equivalent content in language $k \in \mathcal{K}$, we extract hidden states $h_i^{(\ell)} \in \mathbb{R}^d$ from each layer ℓ of the LLM. Specifically, we use the hidden state at the **last token position** of each sentence. We use the Flores Plus dataset [34] with 50 parallel samples across languages (English, Spanish, Russian, Chinese, Hindi) for PCA fitting and 100 for validation.

Language Direction Identification via PCA

We apply Principal Component Analysis (PCA) to the collected hidden states to identify language-specific directions:

1. **Data preparation**: Collect hidden states from the same layer for texts in two languages ℓ_1 and ℓ_2
2. **Centering**: Compute the mean hidden state $\mu = \frac{1}{|H|}\sum_{h \in H} h$ and center the data
3. **PCA**: Compute the principal components of the centered data
4. **Direction extraction**: The first principal component v_1 typically captures the language direction

The language direction v_{lang} satisfies:

$$v_{\text{lang}} = \arg\max_{\|v\|=1} \text{Var}(v^\top H) \tag{6.2}$$

Language Classification

To validate that the identified direction captures language information, we train a linear classifier on the projection of hidden states onto the principal components:

$$\hat{\ell} = \text{sign}(v_{\text{lang}}^\top h + b) \tag{6.3}$$

where b is a learned bias term. Classification accuracy above 95% indicates successful language direction identification.

Activation Steering

During inference, we modify the model's hidden states to reduce code-switching by projecting out the source language component. For each token's hidden state $h_l^{(t)}$ at layer l, we apply:

$$\tilde{h}_l^{(t)} = h_l^{(t)} - s\left(h_l^{(t)} \cdot v^{(l)}\right) v^{(l)} \tag{6.4}$$

The parameter $s \in \mathbb{R}$ controls intervention strength. When $s = 1$, this is an exact orthogonal projection that removes the language component. Values $0 < s < 1$ partially attenuate it, while $|s| > 1$ goes beyond projection removal, actively pushing representations away from the source language direction. In practice, optimal coefficients have $|s| > 1$, indicating that mere removal is insufficient and active counter-steering is needed to overcome the language bias in code-switched input.

Steering is applied only to the final layers where language variance concentrates: for Qwen2.5-1.5B (28 layers), layers 27–28; for Llama-3.2-1B (16 layers), layers 15–16. The optimal steering coefficient s is selected via grid search to minimize KL divergence between the steered model's output distribution and the target language baseline.

Evaluation Metrics

We evaluate steering effectiveness using two complementary evaluation approaches:

Distributional evaluation. We compare the next-token prediction distribution of the steered model against the English baseline using four metrics:

1. **KL divergence** $\mathrm{KL}(P_{\text{EN}} \| P_{\text{steered}})$: lower divergence indicates better mitigation
2. **Jensen-Shannon divergence**: symmetric measure of distributional similarity
3. **Cosine similarity**: between steered and English baseline distributions (higher is better)
4. **Top-50 token overlap**: fraction of shared high-probability tokens with the English baseline

Generation-based evaluation. We generate 100-token continuations and measure code-switching using standard metrics:

1. **Code-Switching Index (CSI)**: fraction of tokens not in the target language (lower is better)
2. **M-Index**: multilingual diversity measure ($0 =$ monolingual, $1 =$ maximally mixed)
3. **I-Index**: ratio of language switch points to token boundaries
4. **Target Language Consistency (TLC)**: $1 -$ CSI (higher is better)

Token-level language identification uses a FastText classifier with a sliding window of 5 tokens.

6.3 Study 3: Complexity-Aware Fine-Tuning

Complexity Estimation

For each training example (x, y), we estimate complexity by running the base model on x and measuring the entropy of predicting y:

$$C(x, y) = \frac{1}{|y|} \sum_{t=1}^{|y|} H(p(\cdot|x, y_{<t})) \tag{6.5}$$

High complexity indicates that the model is uncertain about the correct response, suggesting the example is difficult.

Data Stratification

We partition the training data into three equal complexity tiers by sorting examples by entropy in ascending order:

$$\mathcal{D}_{\text{easy}} = \mathcal{D}[1 : \lfloor N/4 \rfloor] \tag{6.6}$$
$$\mathcal{D}_{\text{medium}} = \mathcal{D}[\lfloor N/4 \rfloor : \lfloor 3N/4 \rfloor] \tag{6.7}$$
$$\mathcal{D}_{\text{hard}} = \mathcal{D}[\lfloor 3N/4 \rfloor + 1 : N] \tag{6.8}$$

The easy and medium examples are combined into a "regular" group $\mathcal{D}_{\text{regular}} = \mathcal{D}_{\text{easy}} \cup \mathcal{D}_{\text{medium}}$ that receives standard SFT, while hard examples receive chain-of-thought distillation.

Training Interventions

Different complexity tiers receive different training treatments:

Easy examples ($C < \tau$): Direct supervised fine-tuning (SFT) with the standard cross-entropy loss:

$$\mathcal{L}_{\text{SFT}}(x, y) = -\sum_{t=1}^{|y|} \log p_\theta(y_t | x, y_{<t}) \tag{6.9}$$

Hard examples ($C \geq \tau$): Chain-of-thought distillation from a larger teacher model $\mathcal{M}_T$:

1. Generate reasoning chains r from the teacher: $r \sim \mathcal{M}_T(\cdot | x, \text{"think step by step"})$
2. Train the student on the augmented examples:

$$\mathcal{L}_{\text{distill}}(x, y) = -\sum_{t=1}^{|r|+|y|} \log p_\theta((r \oplus y)_t | x, (r \oplus y)_{<t}) \tag{6.10}$$

where $r \oplus y$ denotes concatenation of the reasoning chain and the answer.

Training Pipeline

The complete training pipeline consists of:

1. **Complexity scoring**: Run the base model on all training examples to compute complexity scores
2. **Stratification**: Partition data into easy and hard subsets based on threshold τ
3. **Teacher generation**: For hard examples, generate chain-of-thought reasoning from the teacher model
4. **Mixed training**: Fine-tune the student model on the combined dataset with appropriate losses
5. **Evaluation**: Assess accuracy on held-out test data

Hyperparameter Selection

Key hyperparameters include:

- Complexity metric: Single-token answer entropy (selected based on ROC-AUC comparison with alternatives including MASJ reasoning score and cross-entropy)
- Teacher model: An ensemble of large models (DeepSeek-V3-0324, Qwen-3-235B, Llama-4-Maverick) for chain-of-thought generation
- Training schedule: SFT on regular data for 5 epochs, followed by SFT with reasoning-enriched data for hard examples for another 5 epochs
- Fine-tuning method: LoRA with rank 64, alpha 128, dropout 0.05

6.4 Implementation Details

All experiments are implemented in Python using the PyTorch framework [39] and the Hugging Face Transformers library [49]. Hidden state extraction and entropy computation are performed during standard forward passes without requiring gradient computation. Activation steering is implemented using forward hooks that modify hidden states in-place during generation.

Chapter 7
Numerical experiments

This chapter presents the experimental setup and results for each of the three studies comprising this thesis.

7.1 Study 1: Uncertainty Estimation via Token-Wise Entropy

Experimental Setup

Dataset: We use the MMLU-Pro benchmark [46], which contains approximately 12,000 challenging multiple-choice questions with 10 answer choices across 14 diverse domains including Biology, Chemistry, Computer Science, Economics, Engineering, Health, History, Law, Math, Philosophy, Physics, Psychology, Business, and Other.

Models: We evaluate four decoder-only large language models:

- Phi-4 [1]
- Mistral-Small-24B-Base-2501 [22]
- Qwen-2.5-1.5B [51]
- Qwen-2.5-72B [51]

Metrics: We measure the predictive power of entropy features using ROC-AUC for distinguishing correct from incorrect answers, stratified by subject domain and reasoning requirement.

Results

Table 7.1 presents the ROC-AUC scores for predicting incorrect answers using single-token answer entropy across different domains and reasoning complexity levels for all four models.

Figure 7.1 visualizes the ROC-AUC values across subjects for all four models, and Figure 7.2 shows the breakdown by reasoning requirement.

Key findings:

1. Entropy is an effective predictor of answer correctness in knowledge-dependent domains (Biology ROC-AUC 0.83 for Qwen-72B), but performance decreases for reasoning-heavy domains (Math ROC-AUC 0.55 for Phi-4).
2. Larger models show better calibration: Qwen-72B achieves overall ROC-AUC of 0.77 compared to 0.52 for Mistral. Calibration analysis (Figure 7.3) reveals systematic overconfidence across all models, particularly in high-certainty regimes.
3. Low entropy is a better predictor when no reasoning is required (ROC-AUC 0.79 for Qwen-72B) compared to high-reasoning questions (0.73), suggesting entropy primarily captures *knowledge uncertainty* rather than *reasoning difficulty*.

Table 7.1: ROC AUC performance comparison across categories and models using single-token answer entropy.

Category	Phi-4	Mistral	Qwen 1.5B	Qwen 72B
Biology	**0.73**	**0.65**	**0.79**	**0.83**
Economics	0.65	0.58	0.73	0.80
Philosophy	0.65	0.49	0.61	0.74
Physics	0.63	0.44	0.64	0.74
Computer Science	0.61	0.56	0.69	0.78
Psychology	0.61	0.65	0.77	0.77
Chemistry	0.61	0.44	0.64	0.70
Health	0.60	0.55	0.67	0.77
Business	0.58	0.52	0.64	0.74
History	0.58	0.58	0.65	0.67
Engineering	0.58	0.44	0.64	0.73
Other	0.56	0.53	0.66	0.78
Math	0.55	0.54	0.63	0.73
Law	0.50	0.57	0.58	0.69
No Reasoning	**0.59**	**0.59**	**0.72**	**0.79**
Yes Reasoning	0.56	0.52	0.68	0.76
Reasoning (Low)	0.59	0.58	0.71	0.79
Reasoning (Med)	0.57	0.52	0.69	0.76
Reasoning (High)	0.53	0.53	0.61	0.73
Overall	0.58	0.52	0.70	0.77

4. The Model-as-Judge (MASJ) approach performs near-randomly with ROC-AUC values around 0.49–0.57, indicating it captures different aspects of an LLM's confidence that are irrelevant to question complexity.

7.2 Study 2: Language Steering in Latent Space

Experimental Setup

Models: We evaluate two multilingual models:

- Qwen2.5-1.5B (28 layers)
- Llama-3.2-1B (16 layers)

Languages: We test four language pairs: English–Chinese, English–Spanish, English–Russian, and English–Hindi.

Datasets: We use the Flores Plus dataset [34] with 50 parallel samples per language for PCA fitting and 100 for validation. For steering evaluation, we use 2,467 TED Talks samples with artificial code-switching (texts begin in English and switch mid-sentence to the target language).

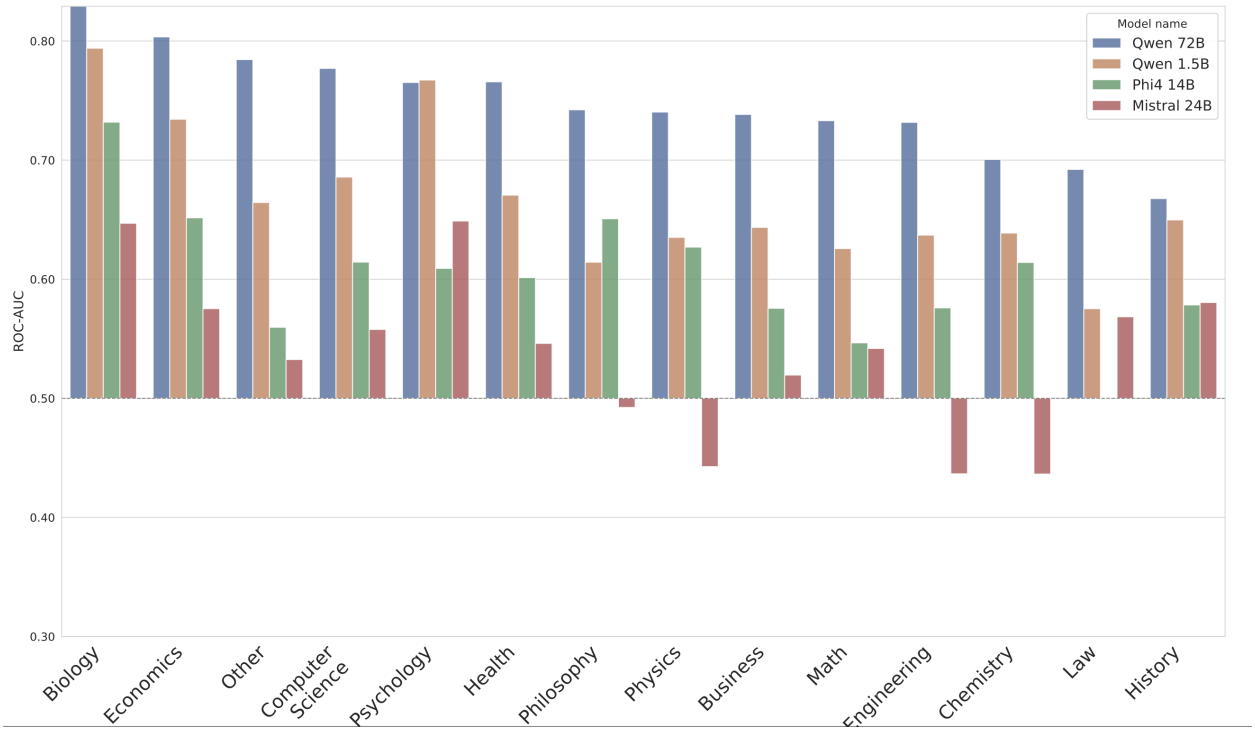


Figure 7.1: ROC-AUC for error prediction by subject for four different LLMs.

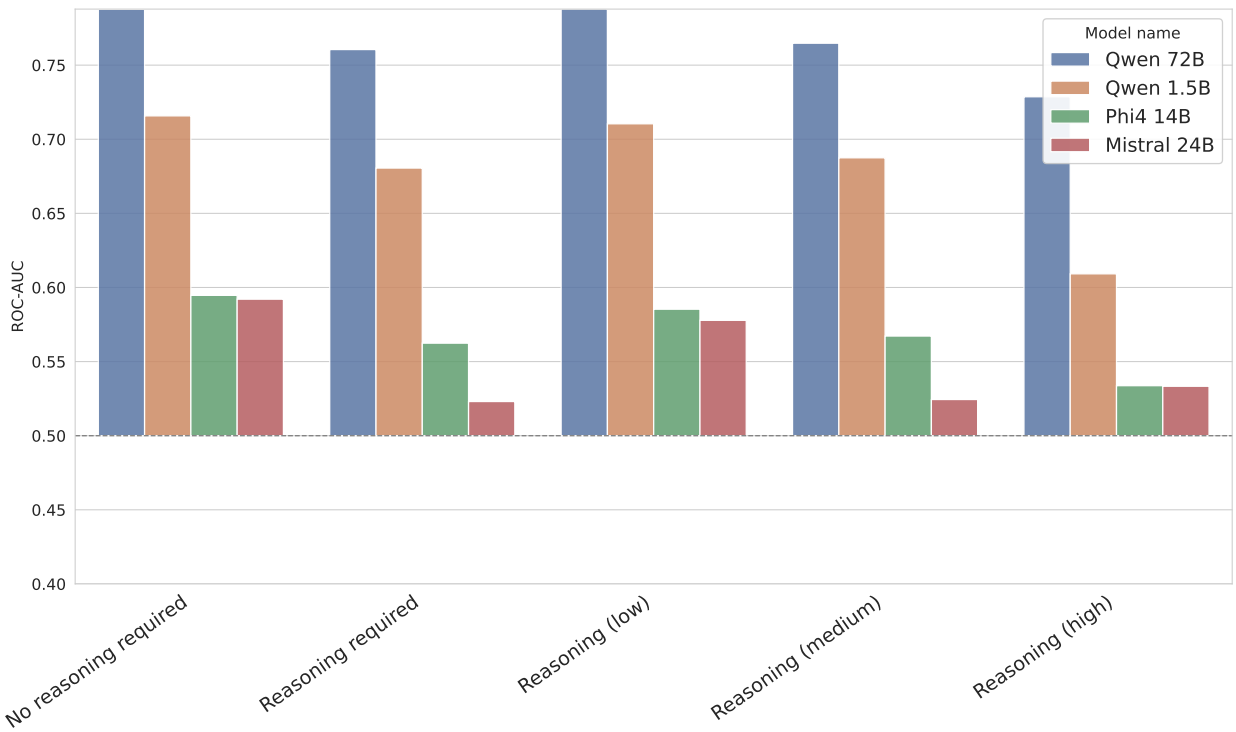


Figure 7.2: ROC-AUC by reasoning requirement for four different LLMs.

Language Direction Identification

Table 7.2 shows the classification accuracy for identifying language from hidden state projections onto the first principal component.

Figure 7.4 shows the final-layer PCA language maps for both models, revealing tight language-specific clusters.

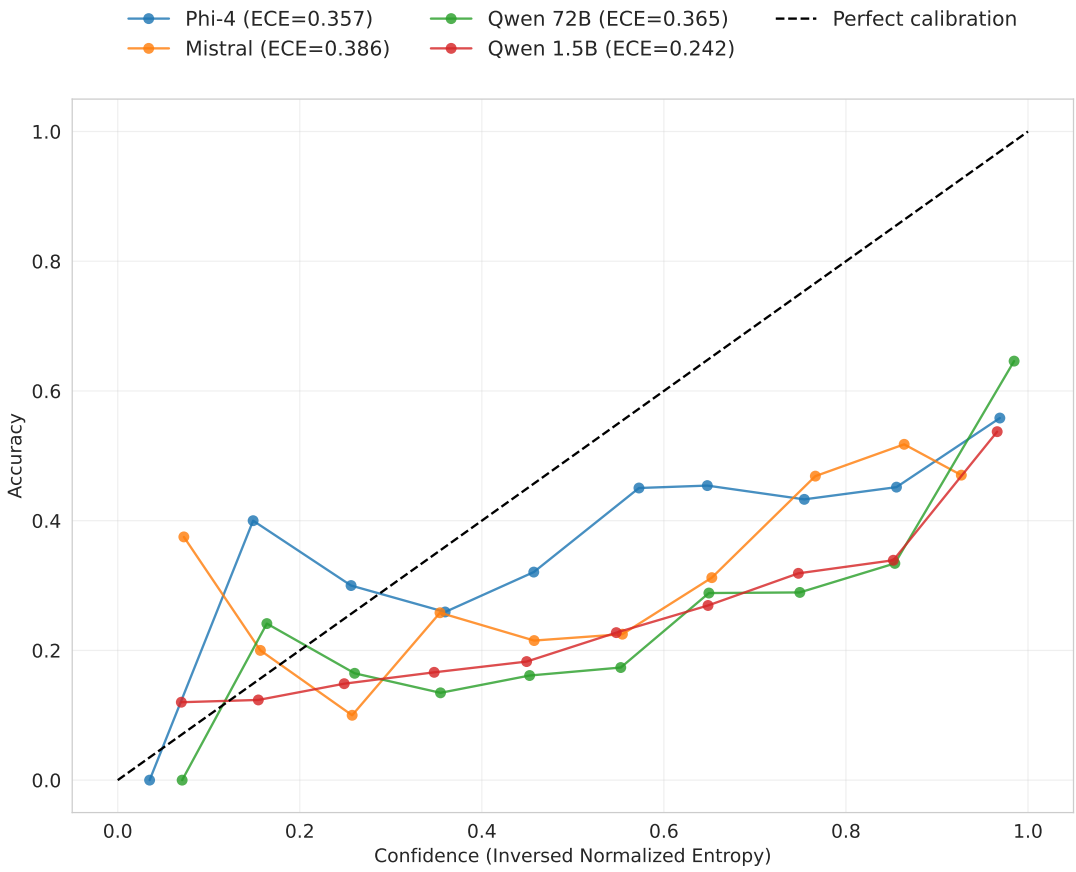


Figure 7.3: Calibration of LLMs based on their entropy. Systematic overconfidence is observed in high-certainty regimes across all models.

Table 7.2: Last layer language classifier accuracy using PCA-identified directions.

Language pair	Qwen	Llama
English – Chinese	0.98	0.98
English – Spanish	0.96	0.96
English – Russian	0.99	0.99
English – Hindi	0.98	1.00

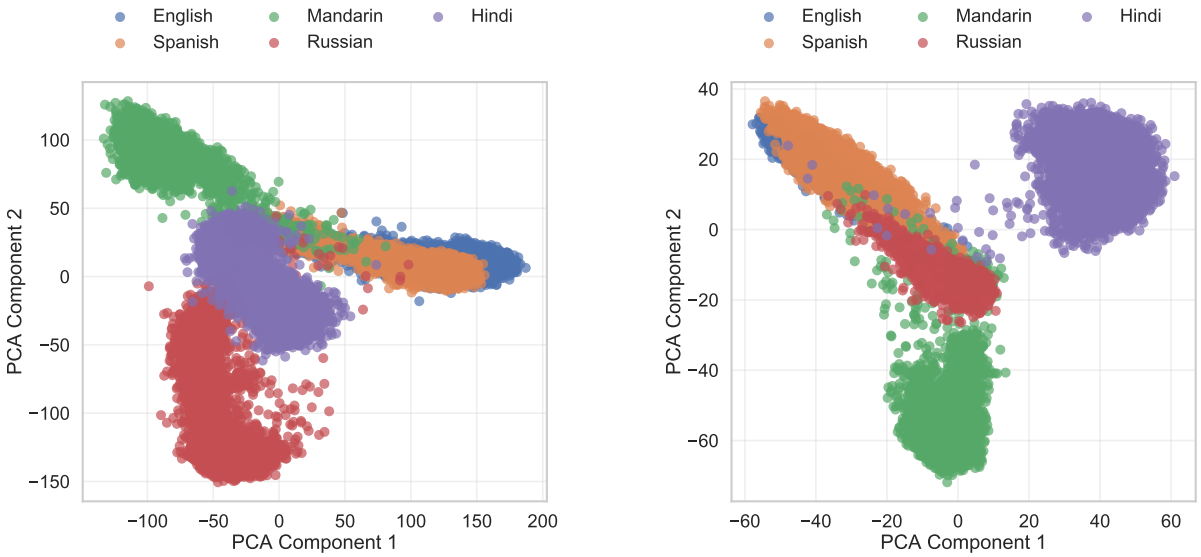
Results show that:

1. Language directions can be reliably identified with 96–100% accuracy using a simple logistic regression classifier on the first principal component.
2. Language identity concentrates in final layers with near-perfect linear separability.
3. English–Spanish shows slightly lower accuracy (0.96) due to these languages being more closely related within the Indo-European family.

Code-Switching Reduction: Distributional Evaluation

We evaluate steering effectiveness by comparing the next-token prediction distribution of the steered model against the English baseline.

Steering reduces KL divergence by 25–55% across language pairs and models (38% average for Qwen, 43% for Llama). The method shows strongest performance for typologically distant



(a) Final layer language map (Qwen)

(b) Final layer language map (Llama)

Figure 7.4: Final-layer embeddings projected onto the first two principal components. Languages organize into tight, language-specific clusters with clear boundaries.

Table 7.3: KL divergence reduction through activation steering (no steering → steering). Lower is better.

Language pair	Qwen	Llama
English – Chinese	10.15 → 4.52	9.05 → 4.77
English – Spanish	7.25 → 4.90	6.52 → 3.48
English – Russian	8.70 → 5.40	7.77 → 3.49
English – Hindi	9.60 → 7.02	8.06 → 6.13

Table 7.4: Cosine similarity between steered and English baseline distributions (no steering → steering, higher is better).

Language pair	Qwen	Llama
English – Chinese	0.02 → 0.22	0.05 → 0.21
English – Spanish	0.12 → 0.18	0.12 → 0.29
English – Russian	0.09 → 0.23	0.10 → 0.31
English – Hindi	0.04 → 0.07	0.06 → 0.16

Table 7.5: Best steering strength coefficient s (KL-optimized via grid search).

Language pair	Qwen	Llama
English – Chinese	-3.2	-5.0
English – Spanish	-2.8	-3.7
English – Russian	-2.2	-3.5
English – Hindi	$+5.0$	-3.6

pairs: English–Chinese achieves 55% KL reduction on Qwen, while English–Russian achieves 55% on Llama. English–Hindi remains the weakest pair (24–27% reduction). Notably, the optimal

coefficient for Qwen English–Hindi is positive ($+5.0$), indicating a reversed PCA direction for this pair, possibly reflecting tokenization asymmetries in the treatment of Devanagari script.

Code-Switching Reduction: Generation-Based Evaluation

We complement the distributional analysis with generation-based evaluation on Llama-3.2-1B. For 500 TED Talks samples per language pair, we generate 100-token continuations and measure code-switching using standard metrics. All steered vs. unsteered differences are significant at $p < 0.001$ (Wilcoxon signed-rank test).

Table 7.6: Generation-based code-switching metrics (Llama-3.2-1B, $n = 500$). English-only baseline: CSI $= 0.02$.

	KL-optimized		CSI-optimized	
Language pair	CSI (↓)	TLC (↑)	CSI (↓)	TLC (↑)
Unsteered (code-switched input, no steering):				
English – Chinese	0.64 / 0.36		—	
English – Spanish	0.62 / 0.38		—	
English – Russian	0.59 / 0.41		—	
English – Hindi	0.81 / 0.19		—	
Steered:				
English – Chinese	0.03	0.97	0.03	0.97
English – Spanish	0.23	0.77	0.11	0.89
English – Russian	0.02	0.98	0.01	0.99
English – Hindi	0.04	0.96	0.01	0.99

With KL-optimized coefficients, CSI drops by 63–95% across language pairs. Target Language Consistency exceeds 0.77 across all pairs, with English–Chinese and English–Russian achieving near-perfect consistency (TLC > 0.97). However, steering coefficients are highly fragile: small changes in magnitude can shift the model from fluent output to degenerate repetition, revealing a sharp trade-off between language control and generation fluency.

7.3 Study 3: Complexity-Aware Fine-Tuning

Experimental Setup

Datasets:

- MMLU-Pro: multiple-choice question answering across 14 domains ($\sim$12,000 examples)
- GSM8K: 8.5K grade school math word problems [7]
- MedMCQA: large-scale medical entrance exam questions [37]

Models:

- Student models: Qwen2.5-3B, Phi-4-Mini, LLama-3.2-3B
- Teacher model ensemble: DeepSeek-V3-0324, Qwen-3-235B, Llama-4-Maverick

Baselines:

- **SFT**: Standard supervised fine-tuning on all data without complexity split
- **Curriculum**: Curriculum-based SFT (3 epochs easy, 3 medium, 4 hard)
- **Distillation**: Full chain-of-thought distillation on all data
- **Alternative**: SFT on hard data, then CoT distillation on easy/medium data

Main Results

Table 7.7 presents fine-tuning results after 10 epochs on MMLU-Pro, showing both accuracy and total tokens processed.

Table 7.7: Accuracy / tokens processed for complexity-aware fine-tuning pipelines after 10 epochs (MMLU-Pro).

Method	Qwen 3B	Phi4-mini
SFT	0.42 / 29k	0.55 / 27k
Curriculum	0.45 / 29k	0.54 / 27k
Distillation	0.49 / 1.97m	**0.63** / 1.51m
Alternative	0.47 / 1.5m	0.58 / 1.2m
Pipeline (ours)	**0.50** / **399k**	0.60 / **268k**

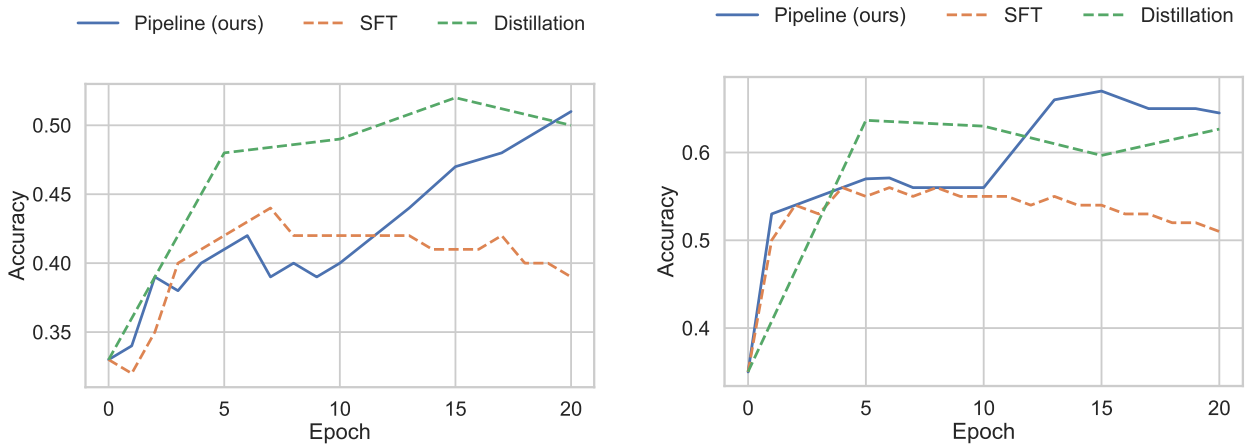
Table 7.8 shows extended results after 20 epochs, including generalization to GSM8K and MedMCQA datasets.

Table 7.8: Accuracy for complexity-aware fine-tuning pipelines after 20 epochs.

	MMLU Pro		GSM8K		MedMCQA	
Method	Qwen 3B	Phi4-mini	LLama 3B	Phi4-mini	LLama 3B	Phi4-mini
SFT	0.39	0.51	0.13	0.19	0.51	0.51
Distillation	0.50	0.63	0.79	0.89	0.56	0.57
Pipeline (ours)	**0.52**	**0.64**	**0.82**	**0.89**	**0.56**	**0.58**

Key findings:

1. Our complexity-aware pipeline matches or outperforms full distillation while using 79–82% less training data (tokens processed).
2. After 20 epochs (Figure 7.5), the pipeline surpasses all baselines: Qwen-3B achieves 0.52 vs. 0.50 (distillation) and 0.39 (SFT); Phi-4-Mini achieves 0.64 vs. 0.63 (distillation) and 0.51 (SFT).
3. The method generalizes across datasets: on GSM8K, LLama-3B achieves 0.82 (pipeline) vs. 0.79 (distillation) and 0.13 (SFT).
4. The pipeline shows a continued uptrend in training accuracy, while full distillation plateaus, suggesting better utilization of training data.



(a) Qwen 3B (MMLU-Pro, 20 epochs)

(b) Phi-4-Mini (MMLU-Pro, 20 epochs)

Figure 7.5: Accuracy dynamics for fine-tuning pipelines over 20 epochs. The complexity-aware pipeline (blue) surpasses full distillation (green dashed) while using 79–82% less training data.

Sensitivity Study

Complexity metric selection. We compare single-token answer entropy against MASJ reasoning score and cross-entropy as complexity metrics. Single-token entropy yields the best ROC-AUC (0.72 for Qwen-3B without domain split) and provides clearer separation between easy and hard examples during SFT training. MASJ reasoning scores achieve ROC-AUC around 0.50–0.57, making them unsuitable for complexity estimation.

Curriculum learning fragility. Standard curriculum learning (easy → medium → hard ordering with SFT) shows extreme sensitivity to hyperparameters. Qwen-3B plateaus at lower accuracy with more training epochs, while Phi-4-Mini quickly overfits, demonstrating that naive complexity-aware ordering without differentiated training strategies is insufficient.

7.4 Summary of Results

Across all three studies, we demonstrate that signals from the model's forward pass provide valuable information for addressing key challenges in LLM deployment:

1. **Uncertainty estimation**: Token-wise entropy achieves ROC-AUC up to 0.83 for predicting answer correctness in knowledge-dependent domains, while MASJ performs near-randomly.

2. **Language control**: PCA-identified language directions enable 25–55% reduction in distributional divergence and 63–95% reduction in Code-Switching Index in generated text across four language pairs.

3. **Training efficiency**: Complexity-aware fine-tuning matches or outperforms full distillation while using 79–82% less training data, generalizing across three datasets and three model architectures.

These results validate our central hypothesis that accessible signals from forward passes encode sufficient information to improve reliability, control, and efficiency without requiring architectural modifications.

Chapter 8
Discussion and conclusion

8.1 Summary of Contributions

This thesis investigated how signals from a single forward pass through large language models—both output distributions and internal representations—can be leveraged to address fundamental challenges in LLM deployment. We presented three interconnected studies that demonstrate the practical utility of this approach.

Study 1: Uncertainty Estimation via Token-Wise Entropy. We demonstrated that single-token answer entropy provides effective signals for predicting answer correctness in knowledge-dependent domains, achieving ROC-AUC scores up to 0.83 on the MMLU-Pro benchmark across four models (Phi-4, Mistral-Small-24B, Qwen-1.5B, Qwen-72B). Entropy primarily captures knowledge uncertainty rather than reasoning difficulty, with performance decreasing for reasoning-heavy domains. The Model-as-Judge approach performs near-randomly (ROC-AUC ≈ 0.49), indicating that different uncertainty estimation methods capture fundamentally different signals. This work was recognized with the Best Paper Award at AINL 2025.

Study 2: Language Steering in Latent Space. We discovered that language-specific directions can be reliably identified in LLM latent space through PCA, achieving 96–100% classification accuracy across four language pairs (English–Chinese, English–Spanish, English–Russian, English–Hindi) on Qwen2.5-1.5B and Llama-3.2-1B. By projecting out the source language component from hidden states during inference, we reduced distributional divergence by 25–55% and achieved 63–95% reduction in Code-Switching Index in generated text, without requiring additional training. This demonstrates that behavioral control can be achieved through geometric manipulation of internal representations, though steering coefficients are fragile and require careful calibration.

Study 3: Complexity-Aware Fine-Tuning. We developed a training pipeline that uses single-token answer entropy to stratify training data by complexity and applies targeted interventions: direct supervised fine-tuning for regular examples and chain-of-thought distillation from a teacher model ensemble for hard examples. This approach achieved accuracy improvements from 0.39 to 0.52 (Qwen-3B) and 0.51 to 0.64 (Phi-4-Mini) on MMLU-Pro while using 79–82% less training data, with results generalizing to GSM8K and MedMCQA datasets across three model architectures.

8.2 Unifying Insights

The three studies share a common finding: the signals produced during a single forward pass through an LLM encode rich information that goes beyond the immediate task of next-token prediction. Specifically:

1. **Output distributions reflect model confidence**: High entropy indicates uncertainty, and the pattern of entropy changes throughout generation correlates with answer correctness.

2. **Internal representations encode interpretable structure**: Language identity is represented

as linear directions in activation space, enabling geometric interventions for behavioral control.

3. **Entropy measures input complexity**: The model's uncertainty about a training example reflects its difficulty, enabling intelligent data stratification.

These insights suggest that LLMs develop internal representations of their own capabilities and limitations that can be accessed and exploited without explicit training.

8.3 Position in the Research Landscape

Our work contributes to several active research areas:

Uncertainty quantification: We extend prior work on semantic entropy [27] by demonstrating that token-wise entropy dynamics provide complementary signals. Our approach requires only a single forward pass, making it practical for real-time applications.

Interpretability and steering: Building on discoveries about linear structure in LLM representations [38, 45], we provide practical methods for language control that require no training and minimal computational overhead.

Efficient training: Our complexity-aware approach relates to curriculum learning [4] and data selection [43], but uniquely combines automatic complexity estimation with targeted training interventions.

8.4 Limitations

Several limitations should be acknowledged:

1. **Model scale**: The uncertainty estimation study uses models from 1.5B to 72B parameters, but the language steering and fine-tuning experiments are limited to small models (1–3B parameters). Generalization to larger models and diverse architectures (e.g., Mixture of Experts) requires further validation.

2. **Language coverage**: While language steering was validated on four language pairs (English–Chinese, English–Spanish, English–Russian, English–Hindi), low-resource languages, right-to-left scripts, and typologically distant languages (agglutinative, tonal) remain unexplored. English–Hindi showed the weakest improvements across all metrics.

3. **Task specificity**: Uncertainty estimation and fine-tuning experiments were conducted on multiple-choice question answering. Performance on open-ended generation, reasoning, and coding tasks needs investigation. The fine-tuning results generalize across MMLU-Pro, GSM8K, and MedMCQA, but all are question-answering tasks.

4. **Steering fragility**: Steering coefficients are highly sensitive—small changes in magnitude can shift the model from fluent output to degenerate repetition. Principled or adaptive coefficient selection methods have not been developed.

5. **Evaluation limitations**: The language steering evaluation uses artificial code-switching (mid-sentence translation), which may not capture the full complexity of naturally occurring code-switching patterns. Human evaluation of semantic preservation and output quality was not conducted.

8.5 Future Directions

This research opens several promising directions:

1. **Combining signals**: The three types of signals (entropy patterns, internal representations, complexity scores) could be combined into unified frameworks for model monitoring and control.

2. **Online adaptation**: The methods could be extended to enable online adaptation during deployment, automatically adjusting model behavior based on observed uncertainty or language patterns.

3. **Multi-dimensional steering**: Beyond language, the geometric structure of latent space could be exploited for steering other attributes such as formality, safety, or domain expertise.

4. **Theoretical foundations**: Developing theoretical understanding of why these signals are predictive would strengthen the foundations for future applications.

5. **Integration with alignment**: The uncertainty and control mechanisms developed here could be integrated with RLHF and other alignment techniques to improve model safety.

8.6 Conclusion

This thesis demonstrated that accessible signals from a single forward pass through large language models provide valuable information for addressing fundamental deployment challenges. Single-token answer entropy predicts answer correctness with ROC-AUC up to 0.83, latent space geometry enables language control with 63–95% reduction in code-switching, and complexity-aware data stratification improves training efficiency while using 79–82% less data. Together, these contributions establish that exploiting forward-pass signals offers a practical path toward more reliable, controllable, and efficient LLMs—without requiring architectural modifications or expensive retraining.

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Innovations

This chapter describes the practical applications and innovation potential of the research results presented in this thesis.

Industrial Applications of Research Results

1. **Real-time reliability monitoring for LLM deployments.** The uncertainty estimation pipeline developed in Study 1 enables automatic detection of unreliable LLM outputs at inference time, requiring only a single forward pass with no additional models or computational overhead. This is directly applicable to high-stakes domains:

   - *Healthcare*: Flagging uncertain diagnostic suggestions before they reach clinicians
   - *Legal*: Identifying potentially hallucinated legal citations or case references
   - *Education*: Warning students when AI-generated explanations may be unreliable
   - *Customer support*: Escalating uncertain responses to human agents

   The finding that entropy-based uncertainty is domain-dependent (effective for knowledge questions, less so for reasoning) provides actionable guidance for deploying such systems: operators can calibrate confidence thresholds per domain.

2. **Multilingual chatbot language control.** The language steering method from Study 2 addresses a concrete industry pain point: multilingual chatbots that unintentionally switch languages mid-response, degrading user experience. The method offers several advantages for industrial deployment:

   - *Zero training cost*: Steering requires only parallel translations for PCA calibration (as few as 50 sentence pairs), with no fine-tuning
   - *Negligible inference overhead*: One dot product and one vector subtraction per token
   - *No model modification*: Works with any decoder-only LLM via forward hooks

   Potential applications include customer service platforms, content localization pipelines, and real-time translation assistants.

3. **Cost-efficient domain adaptation.** The complexity-aware fine-tuning pipeline from Study 3 reduces the computational cost of adapting LLMs to specific domains by 79–82% in terms of training data (tokens processed), while maintaining or improving accuracy. This has direct implications for:

   - *Enterprise AI*: Companies can fine-tune domain-specific models (medical, legal, financial) at a fraction of the current cost
   - *Resource-constrained settings*: Organizations with limited GPU budgets can achieve competitive performance
   - *Rapid iteration*: Faster training cycles enable more frequent model updates as domain knowledge evolves

Open-Source Contributions

All three research projects have released their code and data publicly to facilitate reproducibility and further research:

1. Uncertainty estimation pipeline: `github.com/LabARSS/question-complextiy-estimation`
2. Complexity-aware fine-tuning: `github.com/LabARSS/complexity-aware-fine-tuning`
3. Language steering: `github.com/fxlrnrpt/language-steering-in-latent-space`

Start-up Potential

The combination of the three methods developed in this thesis—uncertainty monitoring, language control, and efficient fine-tuning—forms a coherent product offering for an "LLM reliability and control" platform. Such a platform could provide:

- Automated confidence scoring for LLM outputs (per-domain calibrated)
- Language consistency enforcement for multilingual deployments
- Efficient domain adaptation services with reduced compute requirements

All methods operate on standard transformer architectures without architectural modifications, making them compatible with the most widely deployed open-source LLMs.

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Appendix

A. Prompts for Uncertainty Estimation (Study 1)

The following prompts were used for complexity estimation in the MMLU-Pro evaluation pipeline.

A.1 Single-Token Response Prompts

Standard prompt:

The following are multiple choice questions about subject.
Write down ONLY the NUMBER of the correct answer and nothing else.

With fallback for unknown answers:

The following are multiple choice questions about subject.
If you are certain about the answer return the correct option number, otherwise return 0. Write down ONLY the NUMBER and nothing else.

A.2 Chain-of-Thought Response Prompts

Standard CoT prompt:

The following are multiple choice questions about subject. Explain your thinking process step-by-step. At the end, write down the number of the correct answer by strictly following this format: [number of correct answer].

CoT with fallback:

The following are multiple choice questions about subject. Explain your thinking process step-by-step. At the end, if you are certain about the answer write down the number of the correct answer by strictly following this format: [number of correct answer], otherwise return [0].

A.3 MASJ Prompts

Reasoning requirement estimation:

You are an expert in the topic of the question. Please act as an impartial judge and evaluate the complexity of the multiple-choice question with options below. Begin your evaluation by providing a short explanation. Be as objective as possible. After providing your explanation, you must rate the question complexity by strictly following the criteria: [Number of reasoning steps] – how many reasoning steps do you need to answer this question? Valid answers: low, medium, high.

Education level estimation:

You are an expert in the topic of the question. Please act as an impartial judge and evaluate the complexity of the multiple-choice question with options below. After providing your explanation, you must rate the question complexity by strictly following the scale: "high school and easier", "undergraduate", "graduate", "postgraduate".

B. Training Hyperparameters (Study 3)

Table 8.1: Training hyperparameters for MMLU-Pro experiments.

Parameter	SFT	Pipeline (easy/mid)	Pipeline (hard)	Distillation
Effective batch size	64	64	64	64
Max new tokens	30	30	8192	8192
Optimizer	AdamW	AdamW	AdamW	AdamW
Learning Rate	1e-4	1e-4	1e-4	1e-4

Table 8.2: Training hyperparameters for GSM8K/MedMCQA experiments.

Parameter	SFT	Pipeline (easy/mid)	Pipeline (hard)	Distillation
Effective batch size	256	256	256	256
Max new tokens	100	100	8192	8192
Optimizer	AdamW	AdamW	AdamW	AdamW
Learning Rate	1e-4	1e-4	1e-4	1e-4
LoRA rank	64	64	64	64
LoRA alpha	128	128	128	128
LoRA dropout	0.05	0.05	0.05	0.05

C. Steering Coefficients (Study 2)

Table 8.3: Llama-3.2-1B steering coefficients: KL-optimized (from distributional analysis) vs. CSI-optimized (from generation grid search).

Language pair	KL-opt.	CSI-opt.
English – Chinese	-5.0	-5.0
English – Spanish	-3.7	-5.0
English – Russian	-3.5	-4.0
English – Hindi	-3.6	-5.0

CSI-optimized coefficients tend toward higher magnitudes, which improves language control but increases degeneration risk. The coefficients cluster near the maximum tested value (-5.0), suggesting that direct optimization of generation metrics pushes toward an aggressive regime where language control and fluency trade off sharply.

D. Supplementary Distributional Metrics (Study 2)

Table 8.4: JS divergence (no steering → steering, lower is better).

Language pair	Qwen	Llama
English – Chinese	0.67 → 0.49	0.65 → 0.51
English – Spanish	0.59 → 0.52	0.58 → 0.44
English – Russian	0.62 → 0.50	0.62 → 0.43
English – Hindi	0.64 → 0.60	0.63 → 0.54

Table 8.5: Top-50 token overlap (no steering → steering, higher is better).

Language pair	Qwen	Llama
English – Chinese	3.69% → 26.46%	4.47% → 24.31%
English – Spanish	11.41% → 20.23%	11.45% → 30.74%
English – Russian	7.41% → 25.45%	7.49% → 31.96%
English – Hindi	11.23% → 11.68%	6.54% → 25.88%